

AI Returns Insights

AI-Powered Returns & Cancellation Insights Assistant

1. AI Business Use Case

The objective is to use Generative AI to convert retail returns and cancellation analysis into clear business insights and recommended actions.

The AI assistant uses validated KPIs and analytical findings from the Excel, SQL, and Power BI analysis to:

- Identify major cancellation patterns.
- Highlight products and countries with high cancellation volumes.
- Explain important business trends.
- Generate management-friendly insights.
- Recommend potential business actions.
- Support Business Analysts and business stakeholders in faster decision-making.

The AI is used as an insight-generation and decision-support layer, while the underlying calculations remain based on the SQL, Excel, and Power BI analysis.

2. Business Problem

Retail businesses generate large volumes of transaction data, making it difficult for business stakeholders to quickly identify the reasons and patterns behind cancellations.

The project therefore uses AI to transform analytical results into natural-language business insights so that stakeholders can understand:

- Where cancellations are concentrated.
- Which products have the highest cancellation volume.
- Which countries require attention.
- How cancellation rates change over time.
- What actions the business could investigate.

3. AI Inputs

The AI assistant receives validated outputs from the analytical layer, including:

Total Transactions: ~1M

Cancelled Transactions: ~19K

Cancellation Rate: 1.83%

Cancellation Value: ~1.40M

Highest Cancellation Country: United Kingdom

Highest Cancellation Product: Manual

2009–2010 Cancellation Rate: 2.24%

2010–2011 Cancellation Rate: 1.81%

4. AI Prompt

You are a retail business analyst. Analyze the following validated retail cancellation KPIs and identify the most important business insights. Explain the major cancellation patterns, identify areas requiring attention, and provide practical business recommendations. Do not invent information that is not present in the data. Clearly distinguish between observed facts and possible hypotheses. Keep the analysis suitable for a business stakeholder or management audience.

Data:

Total Transactions: ~1M

Cancelled Transactions: ~19K

Cancellation Rate: 1.83%

Cancellation Value: ~1.40M

Highest Cancellation Country: United Kingdom

Highest Cancellation Product: Manual

Cancellation Rate 2009–2010: 2.24%

Cancellation Rate 2010–2011: 1.81%

5. AI-Generated Business Insights

Based on the validated analytical inputs, the AI assistant generated the following business insights:

1. Overall Cancellation Performance

Approximately 19K cancellations were identified out of around 1M transactions, resulting in an overall cancellation rate of 1.83%.

This indicates that cancellations represent a measurable operational issue and should be monitored regularly.

2. Year-over-Year Improvement

The cancellation rate decreased from 2.24% during 2009–2010 to 1.81% during 2010–2011.

This represents an improvement in cancellation performance over the analyzed period.

3. Country-Level Concentration

The United Kingdom has the highest cancellation volume among the countries analyzed.

This indicates that cancellation-related business processes in the UK market should be investigated further.

4. Product-Level Concentration

The product/category labelled "Manual" has the highest cancellation volume among the products displayed in the Top 10 analysis.

This product/category should be investigated to understand whether cancellations are associated with product demand, ordering processes, inventory availability, or other operational factors.

5. Financial Impact

The analyzed data indicates a cancellation value of approximately 1.40M.

Therefore, cancellation analysis should not focus only on transaction counts. The financial value associated with cancelled transactions should also be considered when prioritizing business problems.

6. Key Business Observation

Cancellations are not evenly distributed across all products and countries. A relatively small number of products and markets account for a significant share of cancellation activity.

This suggests that targeted investigation may be more effective than applying the same corrective action across the entire business.

6. Recommended Business Actions

Based on the identified patterns, the following actions are recommended:

1. Investigate high-cancellation products to identify recurring operational or product-related issues.
2. Perform a focused review of cancellation patterns in the United Kingdom market.
3. Monitor cancellation rate as a regular business KPI through the Power BI dashboard.
4. Prioritize high-value cancellations so that management can focus on areas with greater financial impact.
5. Compare cancellation patterns over time to determine whether corrective actions are improving performance.
6. Use the AI assistant as a decision-support tool while validating important recommendations against the underlying SQL, Excel, and Power BI analysis.

7. Human Validation

AI-generated insights are treated as decision-support outputs rather than final business decisions.

The Business Analyst validates AI-generated observations against the underlying Excel calculations, SQL queries, and Power BI dashboard before presenting them to stakeholders.

This ensures that:

- Numerical values are consistent with the source analysis.
- Business conclusions are supported by available data.
- AI does not introduce unsupported assumptions.
- Recommendations are reviewed by relevant business stakeholders before implementation.

8. AI Limitations

The AI assistant has the following limitations:

- AI-generated explanations may contain assumptions that are not directly supported by the dataset.
- AI cannot determine the actual root cause of a cancellation without additional operational information.
- AI recommendations require validation by business stakeholders.
- The quality of AI insights depends on the quality and completeness of the input data.
- AI should not replace SQL-based calculations or validated business reporting.

Therefore, AI is positioned as an insight-generation and decision-support layer rather than an autonomous decision-making system.

9. Conclusion

The AI component demonstrates how Generative AI can be integrated into a Business Analyst workflow to convert validated analytical results into understandable business insights and actionable recommendations.

The approach combines traditional analysis through Excel, SQL, and Power BI with an AI-based interpretation layer, creating a more efficient and stakeholder-friendly decision-support process.